

Sem 7

Department of Computer Science and Engineering
Unit Test-I, Old Semester 2025
Introduction to GIS and Remote Sensing (OEC-CS702C)

FM: 15

Answer any three questions:

Time Allotted: 45 Minutes
5×3=15

1. Explain in detail about different types of sensors. Elaborate the differences between spatial resolution and spectral resolution.
2. What are the different types of electromagnetic waves used in remote sensing? State the applications of the microwave remote sensing?
3. What are the applications of remote sensing in weather studies? Explain the different types of platforms used in the remote sensing.
4. Explain the different components of remote sensing with details. Explain the challenges and future directions of remote sensing?

4+1 ✓
3+2 ✓
2+3 ✓
3+1+1 ●

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
JGEC/B.TECH/ CSE/OEC-CS 701A/ 2024-25
Human Resources Development and Organizational Behaviour

Full Marks: 15

Times: 45 Minutes

1. What is organizational behavior? Write its importance. What is personality? Write determinants of personality. Write note on attitude and job satisfaction. 3+3+3+3+3

Department of Computer Science and Engineering
Unit Test-I, Odd Semester 2025
Machine Learning (PEC-CS702D)

FM: 15

Time Allotted: 45 Minutes

Answer any three questions:

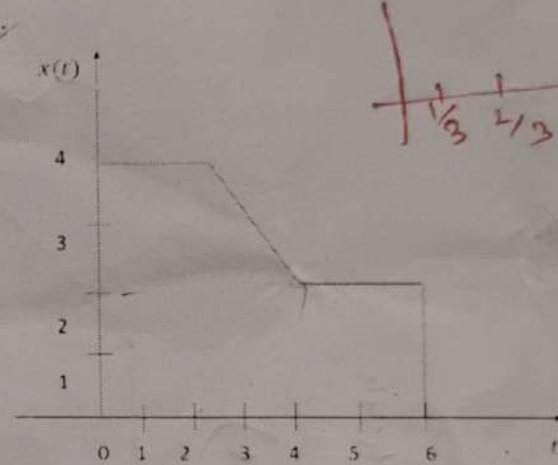
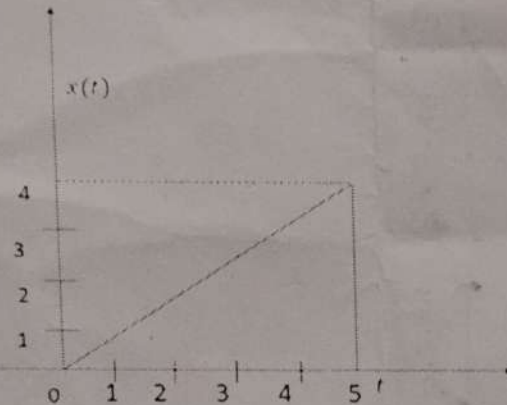
5×3=15

- ✓1 Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters: $A1=(2,10)$, $A2=(2,5)$, $A3=(8,4)$, $B1=(5,8)$, $B2=(7,5)$, $B3=(6,4)$, $C1=(1,2)$, $C2=(4,9)$. Suppose that the initial centres of each cluster are $A1$, $B1$ and $C1$ respectively. Run the k-means algorithm for 2 epochs only. 5
- ✓2 Explain the steps of the K-NN algorithm. Why K-NN is called a lazy learner? Write down the differences between k-means and kernel k-means algorithm. 3+1+1
- ✓3 Explain the PCA algorithm with an example. Why this algorithm is very important in feature engineering? 4+1
- 4 How the SVM algorithm works-explain with an example? Why the kernel trick is used in SVM algorithm? 4+1

1. Sketch and label the even and odd components of the signals shown below:

[5]

Answer any 3 questions

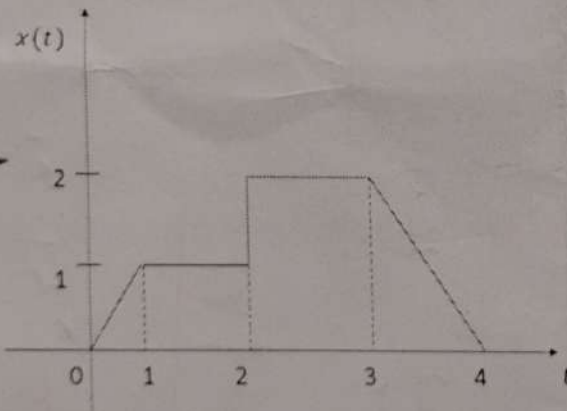


2. Draw the following signals where $x(t)$ is given below:

[5]

- a. $x(3t+4)$
b. $x(3t-4)$

c. $x(3t-4)/9$
d. $x(3t+4)/9$
 $x\left(\frac{3t+4}{9}\right)$



3. Prove

[5]

- i. Even. Even = Even
ii. Odd. Odd = Even

4. Determine whether the signal is Energy or Power:

- i. $x(t) = Ae^{j\omega t}$
ii. $x(t) = tu(t)$

[5]

that following signals with examples:

Advantage of Reinforcement Learning

4x2.5

Signal sp: (10)

even odd power prove

Convolution system with leg

Appli Advan ROC desadv D/B

CLASS TEST-II F.M - 15 SUBJECT: SIGNAL AND SYSTEM (ES-CS701)

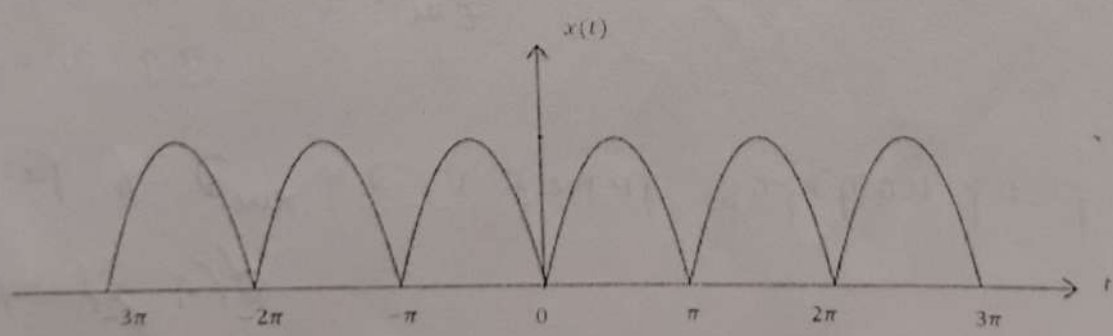
1. i) Draw the Linear Convolution of the following with graphical method only.

$$x(n) = \{1, -2, 3, -2\}; h(n) = \{2, -3, 4\}$$

ii) Draw the circular convolution of the following using graphical method only.
 $x_1(n) = \{1, 3, 5, 7\}$ and $x_2(n) = \{2, 4, 6, 8\}$

OR

2. Determine the continuous time Trigonometric and complex Fourier series and corresponding line and phase spectrum of the following signal.



2-7-16-21-18-8